

Qualification Pack



Fire Safety Supervisor (Oil & Gas)

QP Code: HYC/Q3602

Version: 1.0

NSQF Level: 5

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HYC/Q3602: Fire Safety Supervisor (Oil & Gas)

Brief Job Description

The main responsibility of the fire safety supervisor is to ensure adherence to fire safety systems and procedures and supervise the conduction of firefighting operations in oil and gas industry. He/she develops fire safety systems, procedures, testing and inspection of the firefighting systems and equipment for ensuring workplace safety from fire hazards. The individual is also responsible for inspection & testing of fire safety equipment and coordination with operation and maintenance team for organising safety awareness programs to ensure the safety of the workers and surrounding places. The individual also responsible for developing fire contingency plans.

Personal Attributes

The individual should have a good knowledge and communication skills. He/she must have supervisory skills. The person should be proactive, disciplined, safety-conscious, courageous and a good team player.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HYC/N3603: Introduction to advanced firefighting system including codes & standards](#)
2. [HYC/N3604: Checks, inspection and testing of firefighting equipment for functional reliability](#)
3. [HYC/N3605: Emergency procedures and drills](#)
4. [HYC/N9301: Working Effectively in a team](#)
5. [HYC/N9302: Maintain health, safety and security procedures](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Management of Health, Safety and Environment (HSE)
Country	India

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NSQF Level	5
Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3119.0800
Minimum Educational Qualification & Experience	Completed 3 year diploma after 10th with 1-2 Years of experience OR 12th grade Pass with 3 Years of experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	29/04/2027
NSQC Approval Date	30/04/2024
Version	1.0
Reference code on NQR	QG-05-HY-02524-2024-V1-HSSCI
NQR Version	1.0

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HYC/N3603: Introduction to advanced firefighting system including codes & standards

Description

This unit covers introduction to firefighting systems in line with advanced latest fire safety rules & regulations and procedures specific to oil and gas installations.

Scope

The scope covers the following :

- Carryout safety systems and procedures as per applicable safety rules & regulations

Elements and Performance Criteria

Carryout safety systems and procedures as per applicable safety rules & regulations

To be competent, the user/individual on the job must be able to:

- PC1.** Know and understand Oil Industry Safety Directorate (OISD) standards related to firefighting procedures, systems, equipment and appliances in oil and gas industry
- PC2.** Define Occupational Safety and Health (OSH) policies
- PC3.** Applicable codes & standards related to fire safety
- PC4.** Latest developments in firefighting equipment
- PC5.** Techniques to handle Pressurised Fire Hoses
- PC6.** Hands-on-practice of using Foam making Equipment & different Nozzles
- PC7.** Survival in Smoke and Search & Rescue operation. PPE & Rescue Techniques
- PC8.** Combating Fire inside Oil Storage Tank with Foam System/Equipment
- PC9.** Prevention of Fire/Accidents & Safety Precautions.
- PC10.** Fire-causes, Prevention and Control
- PC11.** Major fire scenarios at site and contingency measures.
- PC12.** Operate firefighting equipment to a greater degree of competence through practice on use of these equipment.
- PC13.** Fighting the road tanker spillage/leakage fire.
- PC14.** Demonstration of BLEVE fire in a pressure vessel.
- PC15.** Combating Fire inside Oil Storage Tank with Foam System/Equipment
- PC16.** Safe work practices while working at height including rescue procedure from height specially during fire incident
- PC17.** Safe work practices during confined space entry job, including buddy system, distinct PTW, JSA specially during fire incident in the confined space
- PC18.** Safe work practices during excavation job including earth collapse
- PC19.** Fire safety measures in high rise buildings
- PC20.** Fire safety measures during transportation of petroleum products through tankers including arresting any leakages

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Fire safety management policy
- KU2.** impact of violation of safety procedures
- KU3.** Knowledge about OISD standards on fire safety
- KU4.** Applicable rules & regulations on fire safety, NFPA guidelines, API guidelines
- KU5.** Latest policies on occupational safety and health
- KU6.** Knowledge about latest developments in firefighting equipments
- KU7.** Knowledge of chemical composition of foam concentrate and its function
- KU8.** Knowledge about pressurized hoses and their handling
- KU9.** Knowledge about fire rescue techniques
- KU10.** Knowledge about different type of fires specially the oil storage tank fires
- KU11.** Knowledge about full body harness, use of SCABA, use of rescue equipment
- KU12.** Knowledge about PTW on confined space entry job
- KU13.** Knowledge about excavation safety and concerned PTW
- KU14.** Knowledge about National Building Code and NFPA
- KU15.** Knowledge about TREM card.
- KU16.** Knowledge of fire detection & suppression system
- KU17.** The concept of Mission Karmayogi
- KU18.** The concept of Swachhata hi Seva.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** keep abreast by reading about new policies at an organization level
- GS2.** communicate effectively in writing
- GS3.** convey information effectively
- GS4.** discuss observations, new policies and schedules and activities
- GS5.** listen with full attention and comprehend the information given by various sources about the site or equipment
- GS6.** use reasoning skills to identify and resolve problems
- GS7.** organize all process manuals so that sorting/ accessing information is easy
- GS8.** analyse information and use logic to address safety-related issues and problems.
- GS9.** determine the safety measures to be followed and how changes in conditions, operations, and the environment will affect outcomes.
- GS10.** undertake and express new ideas and initiatives
- GS11.** Supervise periodically all the fire safety systems

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carryout safety systems and procedures as per applicable safety rules & regulations</i>	20	40	-	-
PC1. Know and understand Oil Industry Safety Directorate (OISD) standards related to firefighting procedures, systems, equipment and appliances in oil and gas industry	1	2	-	-
PC2. Define Occupational Safety and Health (OSH) policies	1	2	-	-
PC3. Applicable codes & standards related to fire safety	1	2	-	-
PC4. Latest developments in firefighting equipment	1	2	-	-
PC5. Techniques to handle Pressurised Fire Hoses	1	2	-	-
PC6. Hands-on-practice of using Foam making Equipment & different Nozzles	1	2	-	-
PC7. Survival in Smoke and Search & Rescue operation. PPE & Rescue Techniques	1	2	-	-
PC8. Combating Fire inside Oil Storage Tank with Foam System/Equipment	1	2	-	-
PC9. Prevention of Fire/Accidents & Safety Precautions.	1	2	-	-
PC10. Fire-causes, Prevention and Control	1	2	-	-
PC11. Major fire scenarios at site and contingency measures.	1	2	-	-
PC12. Operate firefighting equipment to a greater degree of competence through practice on use of these equipment.	1	2	-	-
PC13. Fighting the road tanker spillage/leakage fire.	1	2	-	-
PC14. Demonstration of BLEVE fire in a pressure vessel.	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Combating Fire inside Oil Storage Tank with Foam System/Equipment	1	2	-	-
PC16. Safe work practices while working at height including rescue procedure from height specially during fire incident	1	2	-	-
PC17. Safe work practices during confined space entry job, including buddy system, distinct PTW, JSA specially during fire incident in the confined space	1	2	-	-
PC18. Safe work practices during excavation job including earth collapse	1	2	-	-
PC19. Fire safety measures in high rise buildings	1	2	-	-
PC20. Fire safety measures during transportation of petroleum products through tankers including arresting any leakages	1	2	-	-
NOS Total	20	40	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N3603
NOS Name	Introduction to advanced firefighting system including codes & standards
Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Management of Health, Safety and Environment (HSE)
NSQF Level	5
Credits	3.5
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	29/04/2027
NSQC Clearance Date	30/04/2024

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HYC/N3604: Checks, inspection and testing of firefighting equipment for functional reliability

Description

This unit covers checks, inspection procedure and testing methods of firefighting equipment for their reliability

Scope

The scope covers the following :

- Carryout checks by developing check list, inspection procedures and testing methods

Elements and Performance Criteria

Carryout checks by developing check list, inspection procedures and testing methods

To be competent, the user/individual on the job must be able to:

- PC1.** Checks for functional reliability of firefighting equipment by developing checklist.
- PC2.** Demonstration of BLEVE fire in a pressure vessel.
- PC3.** Donning & maintenance of Breathing Apparatus (BA), search, rescue and firefighting operation in smoke & toxic atmospheres wearing BA
- PC4.** Extinguishing pressure fed & spilled fuel fires
- PC5.** Dealing with electrical and gas fires
- PC6.** Demonstration of OISD Standards (Oil Industry Safety Directorate)
- PC7.** Inspection procedures of fire extinguishers, fire pumps, fire hydrants, monitors, HVLR, Foam pourers.
- PC8.** Testing methods of firefighting equipment
- PC9.** Firefighting & critical operations in vicinity of high heat radiation
- PC10.** Practical Training with SCBA Set in Confined Vessel
- PC11.** Familiarisation of various Fire Fighting Equipment & Systems in Training Ground and demo of BLEVE
- PC12.** Donning & maintenance of Breathing Apparatus (BA), search, rescue and firefighting operation in smoke & toxic atmospheres wearing BA.
- PC13.** Safe escape from smoke-filled areas without the use of Breathing Apparatus (BA)
- PC14.** Search & rescue procedures while wearing in breathing apparatus (BA) under severe smoke Conditions
- PC15.** Dealing with electrical and gas fires
- PC16.** Demonstration of Asset integrity of Fire safety equipments.
- PC17.** Occupational health hazards and remedies

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Knowledge about firefighting equipment and appliances for developing checklist
- KU2.** Knowledge about cascading BA system and SCBA
- KU3.** Basic Knowledge about process safety
- KU4.** Knowledge about testing methods and tools of firefighting equipment
- KU5.** Knowledge about pressure fed & spilled fuel fires
- KU6.** Knowledge about inspection procedure and methodology of firefighting equipment.
- KU7.** Knowledge about firefighting system in different vicinities
- KU8.** Knowledge about SCBA and its purpose
- KU9.** Knowledge about BLEVE
- KU10.** Knowledge about testing methods of different firefighting equipment.
- KU11.** the concept of Mission Karmayogi
- KU12.** the concept of Swacchata hi Seva.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** keep abreast by reading about new inspection techniques
- GS2.** Read about new innovations of testing techniques
- GS3.** convey information effectively about testing techniques
- GS4.** listen with full attention and comprehend the information given by various sources about the site or equipment
- GS5.** use reasoning skills to identify and resolve problems
- GS6.** organize all process manuals so that sorting/ accessing information is easy
- GS7.** analyse information and use logic to address safety-related issues and problems.
- GS8.** determine the safety measures to be followed during testing of firefighting equipment.
- GS9.** undertake and express new ideas and initiatives
- GS10.** Supervise periodically all the fire safety systems

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carryout checks by developing check list, inspection procedures and testing methods</i>	20	30	-	-
PC1. Checks for functional reliability of firefighting equipment by developing checklist.	1	2	-	-
PC2. Demonstration of BLEVE fire in a pressure vessel.	1	2	-	-
PC3. Donning & maintenance of Breathing Apparatus (BA), search, rescue and firefighting operation in smoke & toxic atmospheres wearing BA	1	2	-	-
PC4. Extinguishing pressure fed & spilled fuel fires	1	2	-	-
PC5. Dealing with electrical and gas fires	1	2	-	-
PC6. Demonstration of OISD Standards (Oil Industry Safety Directorate)	1	2	-	-
PC7. Inspection procedures of fire extinguishers, fire pumps, fire hydrants, monitors, HVLRL, Foam pourers.	1	2	-	-
PC8. Testing methods of firefighting equipment	1	2	-	-
PC9. Firefighting & critical operations in vicinity of high heat radiation	1	2	-	-
PC10. Practical Training with SCBA Set in Confined Vessel	1	2	-	-
PC11. Familiarisation of various Fire Fighting Equipment & Systems in Training Ground and demo of BLEVE	2	1	-	-
PC12. Donning & maintenance of Breathing Apparatus (BA), search, rescue and firefighting operation in smoke & toxic atmospheres wearing BA.	1	1	-	-
PC13. Safe escape from smoke-filled areas without the use of Breathing Apparatus (BA)	2	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Search & rescue procedures while wearing in breathing apparatus (BA) under severe smoke Conditions	1	2	-	-
PC15. Dealing with electrical and gas fires	1	1	-	-
PC16. Demonstration of Asset integrity of Fire safety equipments.	2	2	-	-
PC17. Occupational health hazards and remedies	1	1	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N3604
NOS Name	Checks, inspection and testing of firefighting equipment for functional reliability
Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Management of Health, Safety and Environment (HSE)
NSQF Level	5
Credits	4.5
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	29/04/2027
NSQC Clearance Date	30/04/2024

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HYC/N3605: Emergency procedures and drills

Description

This unit covers emergency plans for fire incidents

Scope

The scope covers the following :

- Developing Emergency Response/ Fire contingency plans and checking its efficacy

Elements and Performance Criteria

Emergency procedures and drills

To be competent, the user/individual on the job must be able to:

- PC1.** Developing contingency plan for fire incidents
- PC2.** Preparing Mock drill procedure
- PC3.** Conducting Table Top Exercise
- PC4.** Fire Hydrant Operating Techniques and Communication Procedure
- PC5.** Common First-Aid Techniques, CPR & other Life Saving Practices during Emergency Scenarios
- PC6.** Fire Safety at Installations, Work Permit System & Emergency Response Plan
- PC7.** Use of Portable Fire Extinguishers during Oil & Gas Pipeline Leakage Fire and Electrical Panel Fire
- PC8.** Emergency communication procedure and matrix/ channel
- PC9.** Developing Mock drill schedule
- PC10.** Developing Disaster Management Plan

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge about fire risk analysis and potential fire hazards
- KU2.** Knowledge about fire emergency
- KU3.** impact of violation of safety procedures
- KU4.** Knowledge about OISD standards on fire safety
- KU5.** Applicable rules & regulations on fire safety, NFPA guidelines, API guidelines
- KU6.** Latest policies on occupational safety and health
- KU7.** Knowledge about latest developments in firefighting equipment
- KU8.** Knowledge about oil leakage fires
- KU9.** Knowledge about Work Permit System and the role of fire watcher
- KU10.** Knowledge about CPR

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- KU11.** Knowledge about emergency communication
- KU12.** Knowledge about mock drill techniques.
- KU13.** The concept of Mission Karmayogi
- KU14.** The concept of Swachhata hi Seva.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** keep abreast by reading about new policies at an organization level
- GS2.** communicate effectively in writing
- GS3.** convey information effectively
- GS4.** discuss observations, new policies and schedules and activities
- GS5.** listen with full attention and comprehend the information given by various sources about the site or equipment
- GS6.** use reasoning skills to identify and resolve problems
- GS7.** organize all process manuals so that sorting/ accessing information is easy
- GS8.** analyse information and use logic to address safety-related issues and problems.
- GS9.** determine the safety measures to be followed and how changes in conditions, operations, and the environment will affect outcomes.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Emergency procedures and drills</i>	10	20	-	-
PC1. Developing contingency plan for fire incidents	1	2	-	-
PC2. Preparing Mock drill procedure	1	2	-	-
PC3. Conducting Table Top Exercise	1	2	-	-
PC4. Fire Hydrant Operating Techniques and Communication Procedure	1	2	-	-
PC5. Common First-Aid Techniques, CPR & other Life Saving Practices during Emergency Scenarios	1	2	-	-
PC6. Fire Safety at Installations, Work Permit System & Emergency Response Plan	1	2	-	-
PC7. Use of Portable Fire Extinguishers during Oil & Gas Pipeline Leakage Fire and Electrical Panel Fire	1	2	-	-
PC8. Emergency communication procedure and matrix/ channel	1	2	-	-
PC9. Developing Mock drill schedule	1	2	-	-
PC10. Developing Disaster Management Plan	1	2	-	-
NOS Total	10	20	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N3605
NOS Name	Emergency procedures and drills
Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Management of Health, Safety and Environment (HSE)
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	29/04/2027
NSQC Clearance Date	30/04/2024

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HYC/N9301: Working Effectively in a team

Description

This unit is about working effectively within a team.

Scope

The scope covers the following :

- Effective team work

Elements and Performance Criteria

Effective team work

To be competent, the user/individual on the job must be able to:

- PC1.** maintain clear communication with colleagues
- PC2.** pass on information to colleagues in line with organisational requirements
- PC3.** provide support to the team members
- PC4.** respect the colleagues
- PC5.** fulfil commitments made to colleagues
- PC6.** inform team members timely, if timelines can't be met
- PC7.** take the necessary initiatives to resolve the issues while working in team
- PC8.** adopt gender neutral behaviour while interacting with colleagues
- PC9.** offer assistance to a person with disability (PWD), only if required

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organization policies and procedures related to team performance
- KU2.** the importance of effective communication and establishing good working relationships with colleagues
- KU3.** the importance of creating an environment of trust and mutual respect
- KU4.** the implications of own work on the work and schedule of others
- KU5.** the standard practices in organisation w.r.t communication at various levels
- KU6.** the personal responsibility for completing the task in time
- KU7.** importance of gender equality
- KU8.** importance of showing empathy while interacting with a PwD

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** communicate effectively in writing
- GS2.** read instructions, guidelines/procedures
- GS3.** work in a disciplined manner for meeting commitments and deadline
- GS4.** how to plan and prioritise the work
- GS5.** the importance of consistent and reliable services
- GS6.** apply problem solving approaches in different situations
- GS7.** apply balanced judgments to different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Effective team work</i>	20	30	-	-
PC1. maintain clear communication with colleagues	2	3	-	-
PC2. pass on information to colleagues in line with organisational requirements	2	3	-	-
PC3. provide support to the team members	2	4	-	-
PC4. respect the colleagues	3	4	-	-
PC5. fulfil commitments made to colleagues	2	3	-	-
PC6. inform team members timely, if timelines can't be met	2	4	-	-
PC7. take the necessary initiatives to resolve the issues while working in team	3	4	-	-
PC8. adopt gender neutral behaviour while interacting with colleagues	2	2	-	-
PC9. offer assistance to a person with disability (PWD), only if required	2	3	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9301
NOS Name	Working Effectively in a team
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4
Credits	2
Version	3.0
Last Reviewed Date	16/04/2026
Next Review Date	16/04/2029
NSQC Clearance Date	16/04/2026

Qualification Pack

HYC/N9302: Maintain health, safety and security procedures

Description

This unit is about maintaining health, safety and security procedure at workplace. It covers responsibilities towards self, others, assets and the environment.

Scope

The scope covers the following :

- Follow health and safety measures
- Follow safety procedures during emergency

Elements and Performance Criteria

Follow health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment such as face mask, hand gloves, goggle etc for specific tasks and work conditions
- PC2.** identify the people responsible for maintaining health and safety in the workplace
- PC3.** identify possible causes of risk or accident in the workplace
- PC4.** follow safe working practices while dealing with hazards to ensure the safety of self and others
- PC5.** lift heavy objects safely using correct procedures
- PC6.** follow safety signages
- PC7.** maintain hands hygiene by washing hand frequently and thoroughly with soap and water or alcohol-based hand rub
- PC8.** inform the concerned person of any illness related to self and others
- PC9.** maintain workplace hygiene by disinfecting the equipment and tools regularly

Follow safety procedures during emergency

To be competent, the user/individual on the job must be able to:

- PC10.** respond promptly and appropriately to an accident or in an emergency situation
- PC11.** use appropriate fire extinguishers for different types of fires correctly
- PC12.** follow appropriate rescue techniques during fire hazard
- PC13.** follow good housekeeping practice in order to prevent fire hazards
- PC14.** inform fire safety department about any near-miss incidents in the work place
- PC15.** provide appropriate first aid to victims in an emergency situation
- PC16.** follow the applicable regulations and codes as per safety standard
- PC17.** prepare written accident/incident report and share with the concerned officer/department

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** company's policies on personnel management and duty reporting procedure
- KU2.** reporting structure within organization
- KU3.** health and safety hazards commonly affecting the work environment and related precautions
- KU4.** importance of maintaining personal hygiene using PPE kit, sanitizer and soap
- KU5.** importance of maintaining workplace hygiene
- KU6.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU7.** importance of using protective clothing/equipment while working
- KU8.** various causes of fire
- KU9.** techniques of using different types of fire extinguishers
- KU10.** different materials used for extinguishing fire
- KU11.** various types of safety signs and their significance

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively by writing
- GS2.** read instructions, guidelines/procedures and reports
- GS3.** identify and report potential sources of danger
- GS4.** how to plan the work to meet the deadline
- GS5.** the importance of on time services
- GS6.** apply problem solving approaches in different situations
- GS7.** apply balanced judgments in different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow health and safety measures</i>	9	15	-	-
PC1. use protective clothing/equipment such as face mask, hand gloves, goggle etc for specific tasks and work conditions	1	2	-	-
PC2. identify the people responsible for maintaining health and safety in the workplace	1	-	-	-
PC3. identify possible causes of risk or accident in the workplace	1	2	-	-
PC4. follow safe working practices while dealing with hazards to ensure the safety of self and others	1	2	-	-
PC5. lift heavy objects safely using correct procedures	1	2	-	-
PC6. follow safety signages	1	2	-	-
PC7. maintain hands hygiene by washing hand frequently and thoroughly with soap and water or alcohol-based hand rub	1	2	-	-
PC8. inform the concerned person of any illness related to self and others	1	1	-	-
PC9. maintain workplace hygiene by disinfecting the equipment and tools regularly	1	2	-	-
<i>Follow safety procedures during emergency</i>	11	15	-	-
PC10. respond promptly and appropriately to an accident or in an emergency situation	1	2	-	-
PC11. use appropriate fire extinguishers for different types of fires correctly	2	2	-	-
PC12. follow appropriate rescue techniques during fire hazard	1	2	-	-
PC13. follow good housekeeping practice in order to prevent fire hazards	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. inform fire safety department about any near-miss incidents in the work place	2	2	-	-
PC15. provide appropriate first aid to victims in an emergency situation	1	2	-	-
PC16. follow the applicable regulations and codes as per safety standard	1	2	-	-
PC17. prepare written accident/incident report and share with the concerned officer/department	2	2	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9302
NOS Name	Maintain health, safety and security procedures
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4
Credits	2
Version	3.0
Last Reviewed Date	16/04/2026
Next Review Date	16/04/2029
NSQC Clearance Date	16/04/2026

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training centre (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 70% of aggregate marks to successfully clear the assessment.

Qualification Pack

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HYC/N3603.Introduction to advanced firefighting system including codes & standards	20	40	0	0	60	20
HYC/N3604.Checks, inspection and testing of firefighting equipment for functional reliability	20	30	0	0	50	20
HYC/N3605.Emergency procedures and drills	10	20	0	0	30	20
HYC/N9301.Working Effectively in a team	20	30	-	-	50	15
HYC/N9302.Maintain health, safety and security procedures	20	30	-	-	50	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	110	180	0	0	290	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
MIS	Management Information System
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
MIS	Management Information System
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
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Qualification Pack

NOS	National Occupational Standard(s)
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NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
MIS	Management Information System
NOS	National Occupational Standard(s)
NOS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
MIS	Management Information System
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard Operating Procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding

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GS	Generic Skills
SOP	Standard Operating Procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard Operating Procedure
SS	Stainless Steel
PPE	Personal Protective Equipment

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.